

PowerHive

Presentation of 3° Sprint

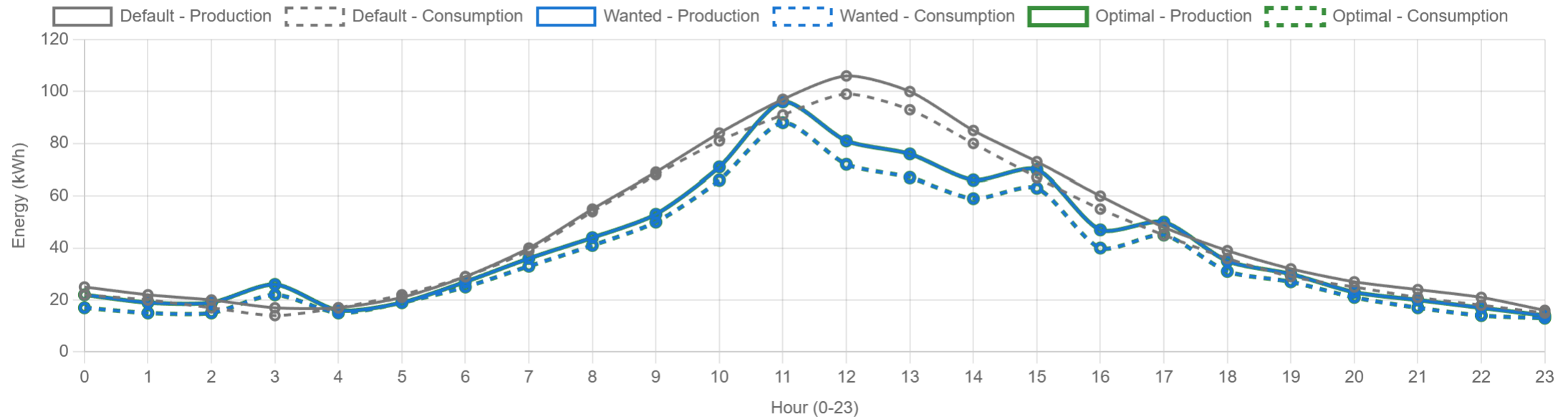
Sprint Goal

Evaluate and optimize battery allocation to maximize shared energy and economic benefits, improve plan management with conflict handling, and enhance analysis history, saving and visualizing past results to enable reuse.

- Refinement of UI/UX and Better Graph Comparisons



Global Production vs Consumption Comparison



- Plan Conflicts Management

The screenshot shows the PowerHive web application interface. At the top, there's a navigation bar with links: Home, Dashboard, Management, Analysis, Ongoing Analysis, History, and Logout. Below the navigation bar, there's a 'Download CSV template' button. A yellow warning box with a triangle icon and the text 'Conflicts detected!' is displayed. The warning message states: 'Some members listed below cannot be saved at this moment due to **conflicts** with your current plan or existing data. Please review the **Status** and **Action** columns for each member to resolve these issues before proceeding with the final save operation.'

Below the warning box, there's a section titled 'Members preview' which contains a table with the following data:

FULL NAME	EMAIL	STATUS	ACTION
Mario Rossi	mario.rossi@example.com	MEMBER ALREADY PRESENT	<button>Solve conflicts</button>
Laura Bianchi	laura.bianchi@example.com	MEMBER ALREADY PRESENT	<button>Solve conflicts</button>
Giuseppe Rudi	laura.bianchi@example.com	EMAIL USED	New mail address required!
Francesco Filocamo	laura.bianchi@example.com	EMAIL USED	New mail address required!
Luca Verdi	luca.verdi@example.com	MEMBER ALREADY PRESENT	<button>Solve conflicts</button>
Stessa Mail di Anna Nery	anna.neri@example.com	MEMBER ALREADY PRESENT	<button>Solve conflicts</button>
Paolo Gialli	paolo.gialli@example.com	MEMBER ALREADY PRESENT	<button>Solve conflicts</button>

- Reuse of previously created energy communities for the next analyses

Add community default from saved Analysis

Tropea

Rosarno

Cassano

Badolato

⚠ Attention: Missing Members

×

The following members from the saved analysis are no longer present in your current plan and were not selected:

- Alice Rossi
- Marco Bianchi

👤 Total Members: 9

✓ Community Default: 3

+ Want to Add: 0

- Want to Remove: 0

MEMBER NAME	CATEGORY	COMMUNITY DEFAULT	WANT TO ADD	WANT TO REMOVE	ACTIONS
👤 Giulia Verdi	PRODUCER	☐	☐	☐	🔍 View
👤 Sara Moretti	PROSUMER	☒	☐	☐	🔍 View

- Batteries Management

[Manage members](#) [Manage batteries](#)

Add New Battery to the Plan

Model Name

Capacity (Kw/h)

Price (€)

[Save Battery](#)

You batteries

MODEL	CAPACITY	PRICE	
Li-Ion C2	25	2999 €	
CR-32	35	4500 €	
Tesla C2	40	5000 €	

- Possibility to run the «Profile Analysis» in Asynchronous Mode

☒	 Chiara De Luca	PRODUCER
☒	 Davide Gallo	CONSUMER
☒	 Elena Romano	PROSUMER
☒	 Federico Conti	CONSUMER
☒	 Giuseppe Rudi	PROSUMER
☒	 Ielpa	PRODUCER

Large number of members selected
You have selected more than 5 members for this analysis. The generation process will be significantly longer and may take several minutes. If you prefer not to wait for the result, you can run the analysis asynchronously by clicking the button below. You will be able to monitor its progress in the **Ongoing Analyses** page.

 Run Asynchronously

Ongoing Analyses

Reload

Analysis #6

Type: 1

Started: 02/12/2025 09:42:14

Num. Members: 9

RUNNING

- Fourth Analysis: Optimal Batteries Assignments

1

Pylontech

Capacity: 10 kWh
Price: **1200 €**

2

Deye RW-M6.1-B

Capacity: 15 kWh
Price: **2000 €**

3

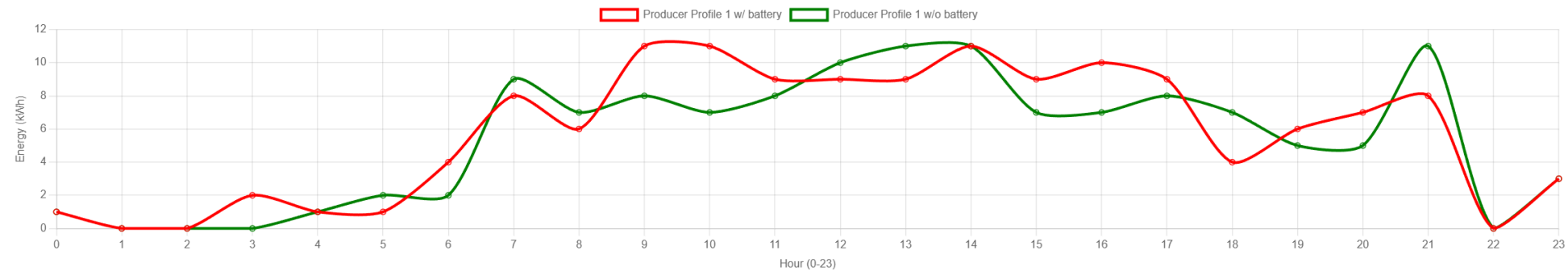
Tesla CF1000

Capacity: 30 kWh
Price: **4500 €**

Optimal Battery Assignment

Mario Rossi  **PRODUCER**

Production Graph



- Fourth Analysis: Overall Financial Analysis

Costs Analysis

Comparison between the costs with/without batteries and visualisation of time to return of investment.

Annual cost without batteries
3,066 € / year

Annual cost with batteries
1,314 € / year

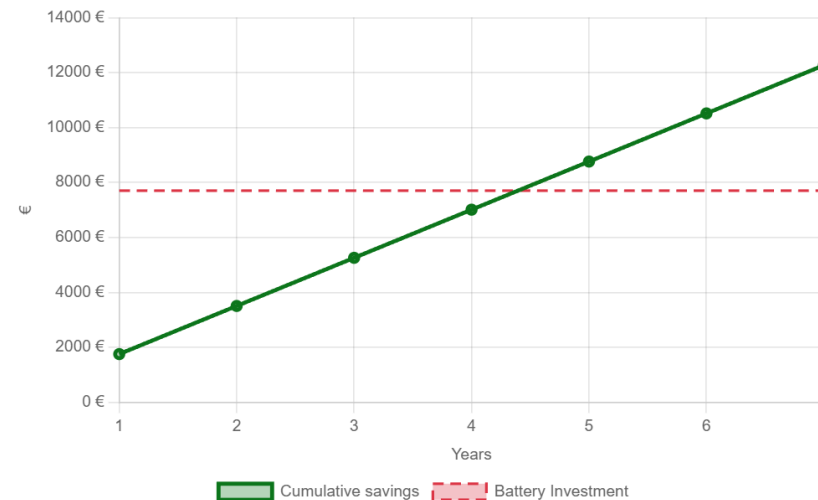
Annual savings thanks to batteries
1,752 € / year

Battery investment
7,700 €

Payback time
4.4 years

Is it convenient to invest in batteries?
YES

Return of Investment



The graph shows the **cumulative savings month by month** compared to the **initial investment** in the battery. The point where the curves intersect is the **estimated payback period**.

- Ability to save the first three types of analyses

Legend: **1** Profile Analysis **2** Optimal Community Formation **3** Community Adjustment Optimization

ANALYSIS NAME	TYPE	CREATION DATE	ACTION
Tropea	2	02/12/2025 09:33:25	Visualize
Rosarno	2	28/11/2025 14:57:19	Visualize
Cassano	2	28/11/2025 14:55:17	Visualize
SanGineto	1	28/11/2025 14:47:27	Visualize
Prova3	3	20/11/2025 17:39:47	Visualize

Next Improvements

- ▶ Possibility to add the asynchronus mode in all analyses
- ▶ Add the possibility to save the «Optimal Battery Assignment» analysis.
- ▶ Reuse the previously saved analyses for the «Optimal Battery Assignment» analysis.
- ▶ General improvements
- ▶ Possibility to choose the batteries and members to perform «Optimal Battery Assignment» analysis.
- ▶ Possibility to use a fixed number of already owned batteries.

Stakeholders requests or ideas ?